

1.6 Classwork: Geometric Growth and Decay — Markscheme

1. [Maximum mark: 6]

The sequence 4, 12, 36, 108, ..., so $u_1 = 4$.

$$(a) \frac{12}{4} = 3, \quad \frac{36}{12} = 3, \quad \frac{108}{36} = 3 \quad \text{M1}$$

$$r = 3 \quad \text{A1}$$

M1 for at least two of the three ratios correctly calculated. A1 for $r = 3$.

A common difference in place of a ratio (e.g. $12 - 4 = 8$) earns no marks.

$$(b) u_n = 4 \times 3^{n-1} \quad \text{A1}$$

Accept any equivalent form, e.g. $\frac{4}{3} \times 3^n$.

$u_n = 4 \times 3^n$ or 12^{n-1} earns no mark.

FT: award A1 for $4 \times r^{n-1}$ with the candidate's own r from (a).

$$(c) u_{10} = 4 \times 3^9 = 78\,732 \quad \text{A1}$$

FT: award A1 for the value obtained by substituting $n = 10$ into the candidate's own expression from (b).

$$(d) 4 \times 3^{n-1} > 1\,000\,000 \quad \text{M1}$$

$$u_{12} = 708\,588 (< 1\,000\,000) \quad \text{and} \quad u_{13} = 2\,125\,764 (> 1\,000\,000)$$

$$n = 13 \quad \text{A1}$$

Accept solving the inequality on the GDC or with logarithms: $n - 1 > 11.31 \dots$, so $n = 13$.

Accept a table search: full marks for showing the adjacent rows u_{12} and u_{13} and stating $n = 13$. A single row ($u_{13} = 2\,125\,764$, so $n = 13$) still earns full marks, with the feedback note that the adjacent row u_{12} is what justifies "smallest". See markscheme-spec §18.

$n = 12.31 \dots$ is not an acceptable final answer — n must be a whole number. Rounding it down to $n = 12$ earns the M1 only.

FT: award A1 for the smallest integer n correctly obtained from the candidate's own expression in (b).

2. [Maximum mark: 7]

Population: 12 000 at the start of 2020, growing 4% per year. Car: \$24 000 at the start of 2020, losing 15% per year.

(a) $r = 1.04$ A1

Accept $1 + 0.04$ or $\frac{104}{100}$.

$r = 0.04$ or $r = 4$ earns no mark.

(b) 2030 is 10 years after 2020: $12\,000 \times 1.04^{10}$ M1

$= 17\,762.9 \dots \approx 17\,763$ A1

Accept 17 800 (3 s.f.), 17 762, 17 762.9 or any value that rounds to 17 760.

An exponent of 9 (giving 17 079.7 ...) or 11 (giving 18 473.4 ...) earns the M1 only.

FT: award M1A1 for $12\,000 \times r^{10}$ evaluated with the candidate's own r from (a), provided $r > 1$.

(c) $r = 0.85$ $24\,000 \times 0.85^5$ M1

$= \$10\,648.93 \approx \$10\,600$ A1

Accept \$10 600 (3 s.f.), \$10 649 or \$10 648.93. M1 is for a multiplier of 0.85 applied five times, however written.

A multiplier of 0.15 (giving \$1.82) or 1.15 earns no marks. An exponent of 4 (giving \$12 528.15) earns the M1 only.

(d) $24\,000 \times 0.85^n < 5000$ M1

After 9 years: \$5558.81 (> 5000); after 10 years: \$4724.99 (< 5000)

10 years A1

Accept solving the inequality on the GDC or with logarithms: $n > 9.65 \dots$, so 10 years.

Accept a table search: full marks for showing both adjacent years and stating 10. A single row (after 10 years the value is \$4724.99, so 10) still earns full marks, with the feedback note that the adjacent year 9 is what justifies "first". See markscheme-spec §18.

9.65 is not an acceptable final answer — the answer must be a whole number of years. Rounding down to 9 earns the M1 only.

FT: award A1 for the smallest whole number of years correctly obtained from the candidate's own multiplier in (c), provided it is between 0 and 1.